Recall

Purpose

A recall object adds a measured or externally simulated time series of water and loads to the model: a point source such as a treatment plant, an inlet from an area outside the model, or, with negative flows, a diversion. The recall output files show what each recall object released into the model.

print.prt row	Files	One row per
recall	recall_day.txt, recall_mon.txt, recall_yr.txt, recall_aa.txt	recall object
basin_psc	basin_psc_day.txt, basin_psc_mon.txt, basin_psc_yr.txt, basin_psc_aa.txt	basin (sum of all recall objects)

Each y in the daily, monthly, yearly or avann column of print.prt opens the matching file. With csvout set to y, a .csv copy of each file is written as well. The basin_psc files are written only when the project has at least one recall object.

Format

- Line 1: the project name from file.cio and the SWAT+ version banner.
- Line 2: column headers.
- Line 3: units.
- Lines 4 and on: data rows.

#	Column	Units	Description
1	jday	-	Day of the year of the last day in the print step.
2	mon	-	Month.
3	day	-	Day of the month.
4	yr	-	Calendar year.
5	name	-	Name of the recall object. In basin_psc files, the project name.
6	type	-	Object type, always recall. basin_psc files have no type column; the header has a blank name in its place.
7	flo	m^3	Water released.
8	sed	tons	Sediment.
9	orgn	kg N	Organic nitrogen.
10	sedp	kg P	Organic (sediment-attached) phosphorus.
11	no3	kg N	Nitrate.
12	solp	kg P	Soluble mineral phosphorus.
13	chla	kg	Chlorophyll a.
14	nh3	kg N	Ammonium.
15	no2	kg N	Nitrite.
16	cbod	kg	Carbonaceous biochemical oxygen demand.
17	dox	kg	Dissolved oxygen.
18	san	tons	Sand.

#	Column	Units	Description
19	`sil`	tons	Silt.
20	`cla`	tons	Clay.
21	`sag`	tons	Small aggregates.
22	`lag`	tons	Large aggregates.
23	`grv`	tons	Gravel.
24	`null`	°C	Water temperature.

The values are the recall record used on the day, unchanged: SWAT+ does not convert the numbers read from the recall time series. The model passes `flo` on to the receiving object as a daily volume in m^3, so that is how the column should be read, although the units row gives m^3/s. The `.csv` copies of the `recall` files use a different units row, which gives `flo` as m^3.

Values by recall time step

The time step of each recall object is set in its recall file:

- Daily: each day uses the record for that day. Outside the years covered by the time series, the values are zero.
- Monthly: the record for the month is used on every day of that month. A monthly row of `recall_mon.txt` is therefore the monthly record multiplied by the number of days in the month. Outside the covered years, the values are zero.
- Yearly: the record for the year is used on every day of that year, so a row of `recall_yr.txt` is 365 or 366 times the yearly record.
- Sub-daily: the daily row is the sum of the records for that day's time steps.

A negative `flo` in a daily recall time series is a withdrawal. The row shows the negative value as read. In addition, SWAT+ removes nitrate, ammonium, nitrite, organic nitrogen, soluble phosphorus and dissolved oxygen from the receiving channel's storage at the channel's current concentration. Those removed amounts are not in the recall files.

Aggregation

Monthly and yearly rows are totals over the step for all columns except `null`, which is a flow-weighted average. The average annual row is the average yearly total. Rows start after the `nyskip` warm-up years, and daily rows follow the daily print window and `interval` in `print.prt`.

`basin_psc` rows are sums over all recall objects, so withdrawals (negative flows) offset releases.

Example

The column names of `recall_day.txt`:

```
jday  mon  day  yr  name  type
flo  sed  orgn  sedp  no3  solp  chla  nh3  no2  cbod  dox  san  sil  cla  sag  lag  grv  null
```

A treatment plant entered as a monthly recall object with `flo` = 1,200 for March shows 1,200 on every March row of `recall_day.txt` and 37,200 in the March row of `recall_mon.txt`.

Related files

- Recall inputs: the recall objects and their time series.
- `print.prt`: the `recall` and `basin_psc` switches.
- Hydrograph outputs: the same column layout for water passed between objects; the `hydout` rows of a recall object show where its water goes.
- Output naming convention: all output files.

Sources

- File opening and headers: `header_write.f90`

- Header types: `hyd_header`, `hyd_header_time`, `rec_header_time`, `hyd_header_units`, `hyd_header_units3` in hydrograph_module.f90
- Writers: recall_output.f90, basin_recall_output.f90
- Daily record selection: `case ("recall")` in command.f90
- Reading the time series: recall_read.f90
- Nutrient removal for withdrawals: recall_nut.f90

Column	Fortran variable
hydrograph (`recall`)	`rec_d(irec) = ob(icmd)%hd(1)`, summed into `rec_m`, `rec_y`, `rec_a`
hydrograph (`basin_psc`)	`brec_d` (sum of `rec_d`), `brec_m`, `brec_y`, `brec_a`
`null`	`hyd_output%temp`